



Bikkad IT Institute

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## 12. SQL CASE Statement

*Compiled by Bikkad IT Institute | Teaching Notes*

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# SQL CASE Statement

12 · CASE Statements — Complete Reference Guide

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## 1. Categorize Data

→ *Task 1: Total sales per category — High (>50), Medium (20-50), Low (<=20)*

```
SELECT
    Category,
    SUM(Sales) AS TotalSales
FROM (
    SELECT
        OrderID,
        Sales,
        CASE
            WHEN Sales > 50 THEN 'High'
            WHEN Sales > 20 THEN 'Medium'
            ELSE 'Low'
        END AS Category
    FROM Sales.Orders
) AS t
GROUP BY Category
ORDER BY TotalSales DESC;
```

## 2. Mapping

→ *Task 2: Map country names to abbreviated country codes*

```
SELECT
```

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```

CustomerID,
FirstName,
LastName,
Country,
CASE
    WHEN Country = 'Germany' THEN 'DE'
    WHEN Country = 'USA'      THEN 'US'
    ELSE 'n/a'
END AS CountryAbbr
FROM Sales.Customers;

```

### 3. Quick Form Syntax

→ Task 3: Abbreviated country codes using both CASE syntaxes

```

SELECT
    CustomerID, FirstName, LastName, Country,
    -- Standard CASE
    CASE
        WHEN Country = 'Germany' THEN 'DE'
        WHEN Country = 'USA'      THEN 'US'
        ELSE 'n/a'
    END AS CountryAbbr,
    -- Quick Form CASE
    CASE Country
        WHEN 'Germany' THEN 'DE'
        WHEN 'USA'      THEN 'US'
        ELSE 'n/a'
    END AS CountryAbbr2
FROM Sales.Customers;

```

### 4. Handling NULLs

→ Task 4: Treat NULL scores as 0 and calculate the average

```

SELECT
    CustomerID,
    LastName,
    Score,
    CASE
        WHEN Score IS NULL THEN 0
        ELSE Score
    END AS ScoreClean,
    AVG(
        CASE WHEN Score IS NULL THEN 0 ELSE Score END
    ) OVER () AS AvgCustomerClean,
    AVG(Score) OVER () AS AvgCustomer

```

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FROM Sales.Customers;

## 5. Conditional Aggregation

→ *Task 5: Count orders per customer where Sales > 30*

```
SELECT
    CustomerID,
    SUM(
        CASE WHEN Sales > 30 THEN 1 ELSE 0 END
    ) AS TotalOrdersHighSales,
    COUNT(*) AS TotalOrders
FROM Sales.Orders
GROUP BY CustomerID;
```

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